

Surgical interruption of pelvic nerve pathways in dysmenorrhea: a systematic review of effectiveness.



OBJECTIVES: To assess the effectiveness of surgical interruption of pelvic nerve pathways in primary and secondary dysmenorrhea. **Data sources.** The Cochrane Menstrual Disorders and Subfertility Group Trials Register (9 June 2004), CENTRAL (The Cochrane Library, Issue 2, 2004), MEDLINE (1966 to Nov. 2003), EMBASE (1980 to Nov. 2003), CINAHL (1982 to Oct. 2003), MetaRegister of Controlled Trials, the citation lists of review articles and included trials, and contact with the corresponding author of each included trial. **REVIEW METHODS:** The inclusion criteria were randomized controlled trials of uterosacral nerve ablation or presacral neurectomy (both open and laparoscopic procedures) for the treatment of dysmenorrhea. The main outcome measures were pain relief and adverse effects. Two reviewers extracted data on characteristics of the study quality and the population, intervention, and outcome independently. **RESULTS:** Nine randomized controlled trials were included in the systematic review. There were two trials with open presacral neurectomy; all other trials used laparoscopic techniques. For the treatment of primary dysmenorrhea, laparoscopic uterosacral nerve ablation at 12 months was better when compared to a control or no treatment (OR 6.12; 95% CI 1.78-21.03). The comparison of laparoscopic uterosacral nerve ablation with presacral neurectomy for primary dysmenorrhea showed that at 12 months follow-up, presacral neurectomy was more effective (OR 0.10; 95% CI 0.03-0.32). In secondary dysmenorrhea, along with laparoscopic surgical treatment of endometriosis, the addition of laparoscopic uterosacral nerve ablation did not improve the pain relief (OR 0.77; 95% CI 0.43-1.39), while presacral neurectomy did (OR 3.14; 95% CI 1.59-6.21). Adverse events were more common for presacral neurectomy than procedures without presacral neurectomy (OR 14.6; 95% CI 5-42.5). **CONCLUSION:** The evidence for nerve interruption in the management of dysmenorrhea is limited. Methodologically sound and sufficiently powered randomized controlled trials are needed.